MLOps Lab Report

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GitHub Repository

URL: https://github.com/OMAR-AmD/fraud-mlops

Git History Verification:

```
73fdce6 Starter project
501eb15 Initialise DVC
0e312dc Data v1
92a9e8d Model on data v1
dc93f9a Add data test: unique transaction_id
563be70 Data v2
af73fa9 (HEAD -> main, origin/main) Retune for data v2
```

1. Data Versioning & Time Travel (Step 8.7)

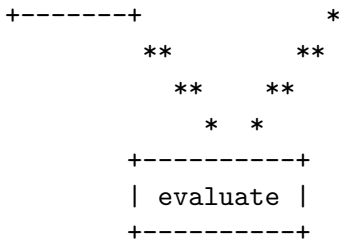
To travel back to the first dataset, the specific commit ID for "Data v1" (`0e312dc`) was located using `git log --oneline`.

- Command run to move the Git ticket: `git switch --detach 0e312dc`.
- Command run to restore the old data file: `dvc checkout`.
- When executing the Python hash check, the fingerprint returned (`758A5A69A84269EBAE16B70AAA1058EA`) exactly matched the original Data v1 fingerprint stored in the `transactions.csv.dvc` ticket.
- *Note:* Git only moves the ticket; the actual data file only reverts when `dvc checkout` is executed. When returning to `main`, the hash successfully updated to Data v2 (`72AA507598DD4F36A7B178E83`

2. Pipeline Architecture & Rerun Logic

Based on the `dvc dag` output, the pipeline is structured as follows:





- **Data or model settings changed (max_depth, n_estimators, seed):** Both the `train` and `evaluate` stages will rerun.
- **Only a gate changed (min_recall, min_precision):** Only the `evaluate` stage reruns.
- **Nothing changed:** Nothing reruns (verified by `Data and pipelines are up to date`).

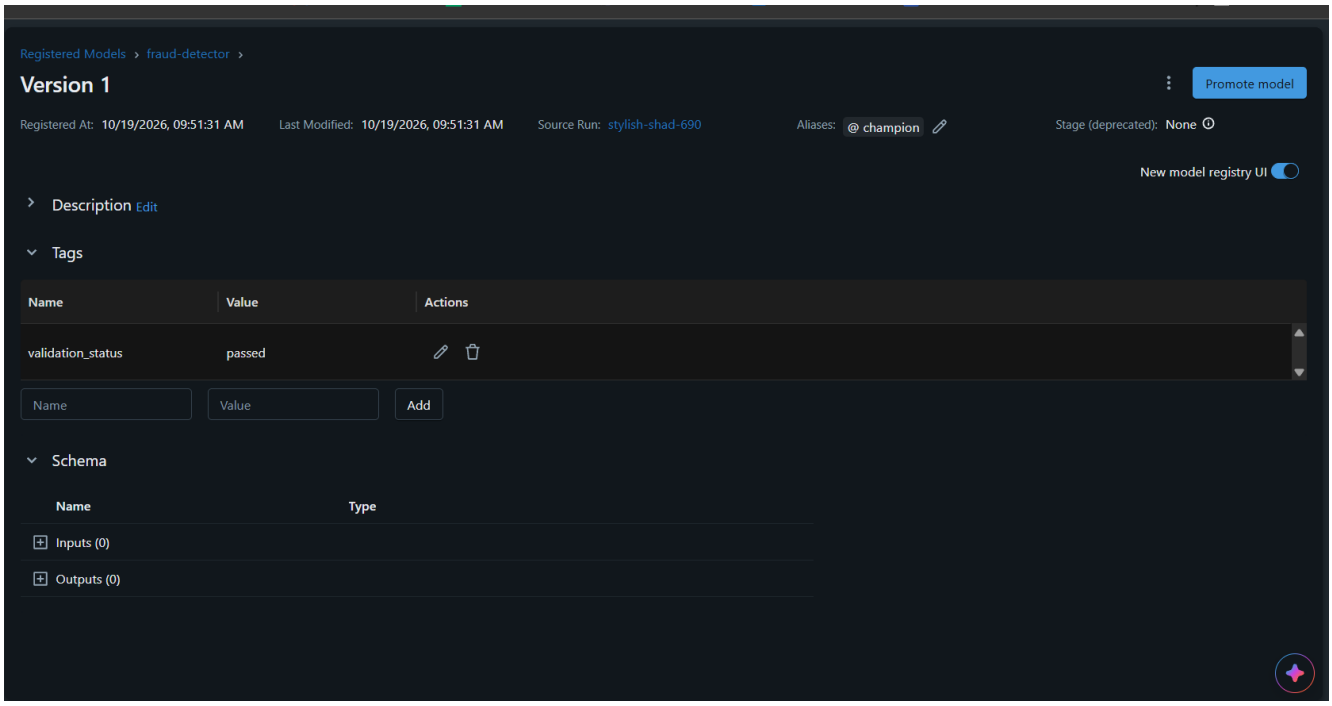
3. Checkpoint Answers

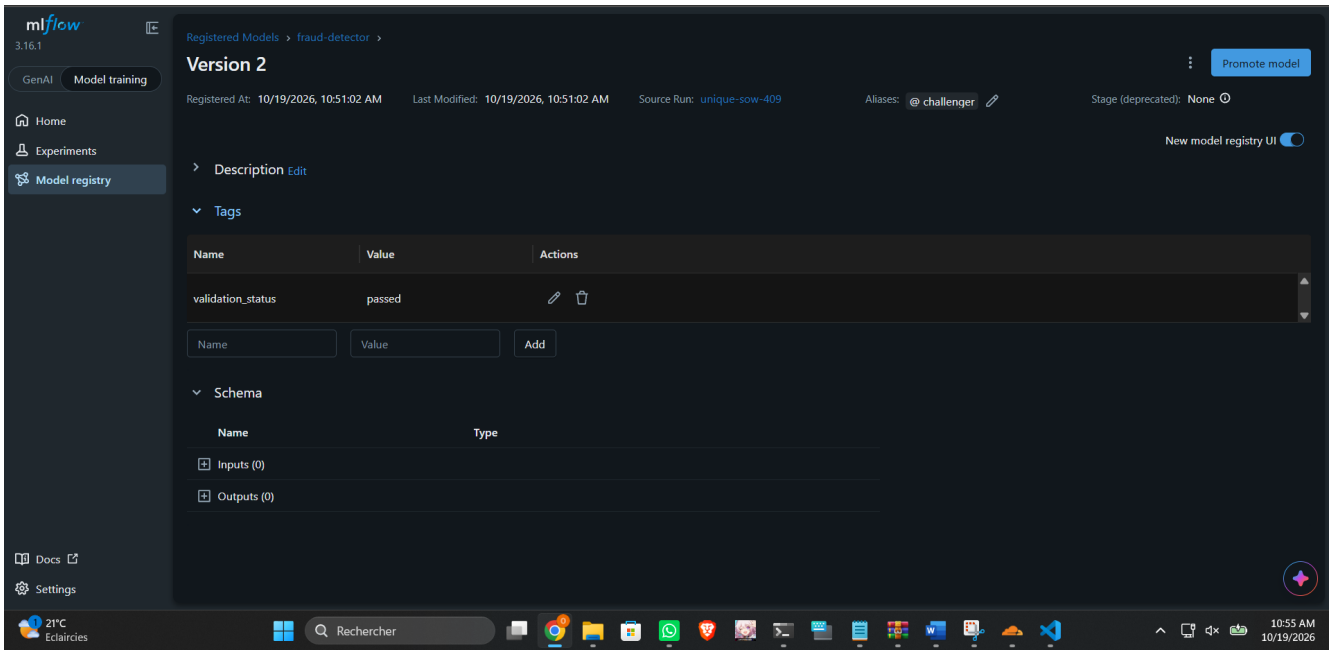
- **Step 1.5:** The library versions are fixed in `requirements.txt` to ensure the virtual environment uses the exact tested versions, preventing clashes. This guarantees that if a classmate installs the project, they will generate the exact same model.
- **Step 3.7:** Git only stores the data's fingerprint (MD5 hash), its size, and the file path. The actual data file is never stored in Git.
- **Step 5.4:**
  - Git Commit ID: `92a9e8d` (Model on data v1)
  - Data MD5 Fingerprint: `758A5A69A84269EBAE16B70AAA1058EA`
- **Step 8.8:** Retuning the model for data v2 was legitimate to adapt to the new information. Someone cannot simply bypass a failed gate by lowering the minimum threshold in `params.yaml` because changes are blocked by pull requests, mandatory code reviews, and GitHub CI branch protection rules.

4. Required Screenshots

Instructions: Replace the placeholder files below with your actual screenshot files in your LaTeX editor.

MLflow Registry: Promotion and Rollback





GitHub CI: Red Pull Request

